

Getting Started with Statistics

Chapter 1: Introduction

Sections 1.1 and 1.2

MAT 1272

Introduction to Statistics

Based on the course textbook

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Introductory Statistics, 10th Edition

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What do you think of when you hear the word **statistics**?

numbers

graphs

probability

data

surveys

decisions

Today we will turn these everyday ideas into a precise mathematical language.

- ① What does the word **statistics** mean?
- ② What are the two major types of statistics?
- ③ What is the difference between a **population** and a **sample**?
- ④ What are an **element**, **variable**, **observation**, and **data set**?
- ⑤ How do these ideas fit together?

The big idea

Statistics is not just “doing calculations.” It is a way of learning from data and making decisions under uncertainty.

The textbook uses the word **statistics** in two different ways.

Meaning 1: statistics

A collection of numerical facts or values.

Examples:

- age of a student
- family income
- percentage of respondents
- number of airline arrivals

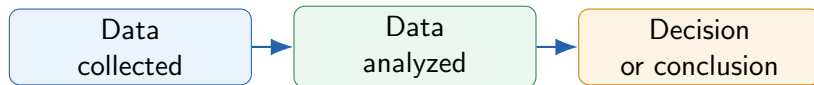
Meaning 2: Statistics

The field of study concerned with collecting, analyzing, presenting, and interpreting data—and making decisions based on those analyses.

Notice the distinction

A *statistic* can be a number. *Statistics* is also the discipline that tells us what to do with data.

Every day we make decisions with incomplete information.



- Statistics helps us organize information.
- Statistics helps us recognize patterns.
- Statistics helps us quantify uncertainty.
- Statistics helps us make **educated** decisions rather than guesses.

Statistics is a **process**.

- 1 Ask a question.
- 2 Collect data.
- 3 Organize and display the data.
- 4 Analyze the data.
- 5 Interpret the results.
- 6 Make a decision or conclusion.

Keep this picture in mind

We will repeatedly move from **data** → **information** → **conclusions**.

Descriptive statistics

Describe what the data say.

- organize
- display
- summarize

Examples:

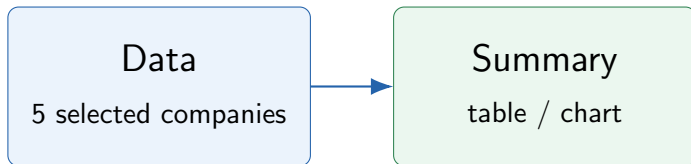
- tables
- graphs
- averages
- percentages

Inferential statistics

Use a sample to learn about a population.

- estimate
- predict
- make decisions

The key word is **generalize**.



The textbook's lobbying-spending example reports how much five selected companies spent in 2014.

Why descriptive?

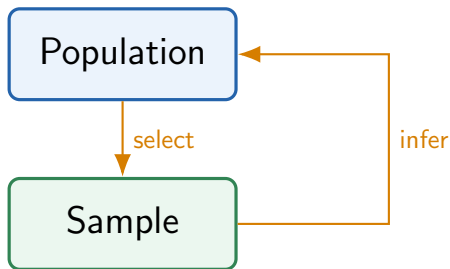
The chart summarizes the data that were collected. It does not use those five companies to make a prediction about all companies.

Company	2014 lobbying spending
AT&T	\$14.2 million
Comcast	\$17.0 million
Google	\$16.8 million
Time Warner Cable	\$7.8 million
Verizon	\$13.3 million

Ask yourself

What can we legitimately say from this table?

- We can compare the five companies.
- We can identify the largest and smallest values.
- We can calculate summaries such as the mean.
- We should not automatically claim that these five represent every company.



The central question

How can information from a **sample** tell us something about a larger **population**?

This is where uncertainty enters. Probability will become an important tool for quantifying that uncertainty.

The textbook gives a poll of 176,903 American adults.

54.1%
thriving

42.1%
struggling

3.8%
suffering

The important statistical move is not the percentages themselves. It is the idea of using the results of a **sample** to make statements about a broader **population**.

Descriptive

What do the
data show?

Inferential

What can we conclude
about a population?

Quick check

If I make a bar graph of the exam scores of the students in *this class*, am I doing descriptive or inferential statistics?

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Descriptive!

Now we build the vocabulary that will appear throughout the course.

Element

Variable

Observation

Data Set

Population

Sample

Why vocabulary matters

Once these words are precise, statistical questions become much easier to describe and solve.

Definition

An **element** (or member) is a specific subject or object about which information is collected.

Examples of elements:

- a person
- a student
- a company
- a household
- a state or country

Think of an element as...

one “row” or one individual unit of interest.

Definition

A **variable** is a characteristic under study that can take different values for different elements.

Examples:

- household income
- age
- number of courses taken
- total wealth
- number of houses built in a city

Key phrase

A variable is something that can **vary**.

Suppose our elements are the eight people listed in the textbook's table.

Name	Total wealth (billions of dollars)
Bill Gates	79.2
Carlos Slim Helu	77.1
Warren Buffett	72.7
Amancio Ortega	64.5
Larry Ellison	54.3
Charles Koch	42.9
David Koch	42.9
Christy Walton	41.7

- **Element:** one person, such as Warren Buffett.
- **Variable:** total wealth.

Definition

The value of a variable for a particular element is called an **observation** or **measurement**.

Using the previous table:



Therefore

For Warren Buffett, \$72.7 billion is one **observation** of the variable “total wealth.”

Definition

A **data set** is a collection of observations on one or more variables.

Think of a data set as a collection of answers to our questions.

Observation 1

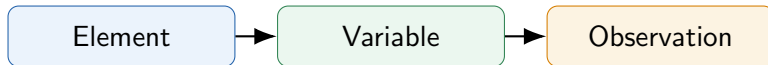
Observation 2

Observation 3

Observation 4



Data Set



One complete sentence

For the element **Warren Buffett**, the variable **total wealth** has the observation **72.7 billion dollars**.

Then

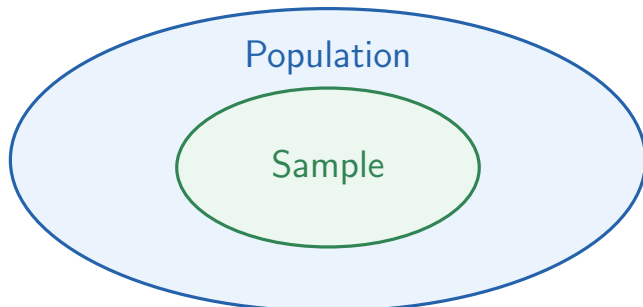
Collect many observations → obtain a **data set**.

Population

The collection of **all elements of interest**.

Sample

A **subset** of the elements selected from the population.



Suppose we want to know:

“What do City Tech students think about ...?”

Population

All City Tech students who are part of the group we want to study.

Sample

The students we actually select and survey.

Statistical goal

Use the sample to learn something about the population— while recognizing that the sample is not the entire population.

For each situation, decide which type of statistics is being used.

- ① A professor calculates the average exam score for the 28 students in her class.
- ② A researcher surveys 500 voters and uses the results to predict the opinion of all voters in the city.
- ③ A company makes a pie chart showing the percentage of its employees in each department.
- ④ A scientist uses a sample of trees to estimate the average height of all trees in a forest.

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Situation	Type
1	Descriptive
2	Inferential
3	Descriptive
4	Inferential

A college collects the following information from 100 students:

Student	Number of credits completed
Student A	32
Student B	45
Student C	28
Student D	61
$\vdots$	$\vdots$

Identify:

- 1 an element
- 2 the variable
- 3 one observation
- 4 the data set

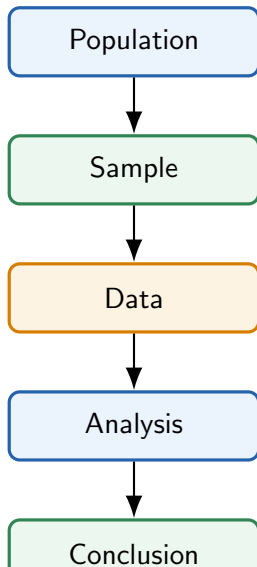
- **Element:** one student, such as Student B.
- **Variable:** number of credits completed.
- **Observation:** 45 credits for Student B.
- **Data set:** the collection of the 100 observations.

A useful mental model

Who? → Element

What characteristic? → Variable

What value? → Observation



- ① Statistics has both a **numerical** meaning and a **discipline** meaning.
- ② **Descriptive statistics** organize and summarize data.
- ③ **Inferential statistics** use sample information to make decisions or predictions about a population.
- ④ An **element** is the individual unit we study.
- ⑤ A **variable** is a characteristic that can vary.
- ⑥ An **observation** is a value of a variable for one element.
- ⑦ A **data set** is a collection of observations.
- ⑧ A **sample** is selected from a **population**.

Suppose you want to know the average amount of time City Tech students spend commuting to campus.

What are the:

Population?

Sample?

Variable?

Think before we move on

Who are we interested in? Who do we actually measure? What characteristic are we measuring?

We now know the language.

Next we will learn how to organize and display data.

